SINGAPORE INSTITUTE OF TECHNOLOGY

(INF2009) Edge Computing & Analytics

Project Briefing

Project Teams Formation

- **Project Team Formation**
  - Form groups of 4 or 5 pax (per group) for the team project.
  - Go to **Class Activities > Groups** on xSiTe and enroll yourself.
    - Deadline for enrollment: **9th January 2026.**
  - You **may** form groups across lab sessions.
  - If you do not enroll by the above date, you will be automatically assigned a group.

Project Proposal

- Submit the project proposal by **16th January 2026** on xSiTe using the template provided.
  - Include a list of equipment required in the proposal.
- What specific issue or problem has been identified that requires a solution?
- What is the potential impact of addressing this challenge?
- What are the main goals of this project or research?
- What specific outcomes are expected by the end of this study?
- Equipment Request: Submit a document with the list of equipment required on LMS Dropbox along with the Proposal

Design & Justification – Deadlines

- Design & Justification **(Week 8: Friday, 27th Feb @ 2359hrs)**
  - **5-Slides + 6-min Video**
    - Introduction, Problem Statement and Objectives
    - Design (System Architecture only)
    - Justification

Design & Justifications

Problem Statement

- Clarity and Relevance:
  - Does the problem statement clearly define the issue that the project addresses?
  - How relevant is the problem in the context of edge computing and analysis?
- Complexity and Specificity:
  - Does the problem statement identify specific challenges or complexities inherent in edge computing environments?
  - How well does it articulate the nuances of the project's focus area?
- Originality and Innovation:
  - Is the problem statement original, proposing a unique challenge or perspective within the field?
  - Does it suggest innovative approaches or solutions?

Design

- Architectural Appropriateness:
  - How suitable is the chosen architecture for edge computing scenarios?
  - Does the design efficiently handle computing at the edge, including data processing and storage?
- Integration with Existing Technologies:
  - How well does the design integrate with existing edge computing technologies and frameworks?
  - Is there compatibility with standard protocols and interfaces?
- Resource Optimization :
  - Does the design consider resource constraints typical in edge computing, like limited bandwidth or computational power?
  - How effectively does it optimize the use of these resources?
- Scalability and Flexibility:
  - Is the design scalable and flexible to accommodate varying workloads and future expansions?
  - Can it adapt to different edge computing environments and use cases?

Justification

- Feasibility and Practicality:
  - How feasible is the implementation of the proposed design?
  - Is the approach practical for real-world edge computing applications?
- Impact and Benefits:
  - What are the potential impacts and benefits of the project in the context of edge computing?
  - How does it contribute to advancing the field or solving pertinent issues?
- Risk Assessment and Mitigation:
  - Are potential risks and challenges identified and addressed?
  - Does the project include strategies for mitigating these risks?

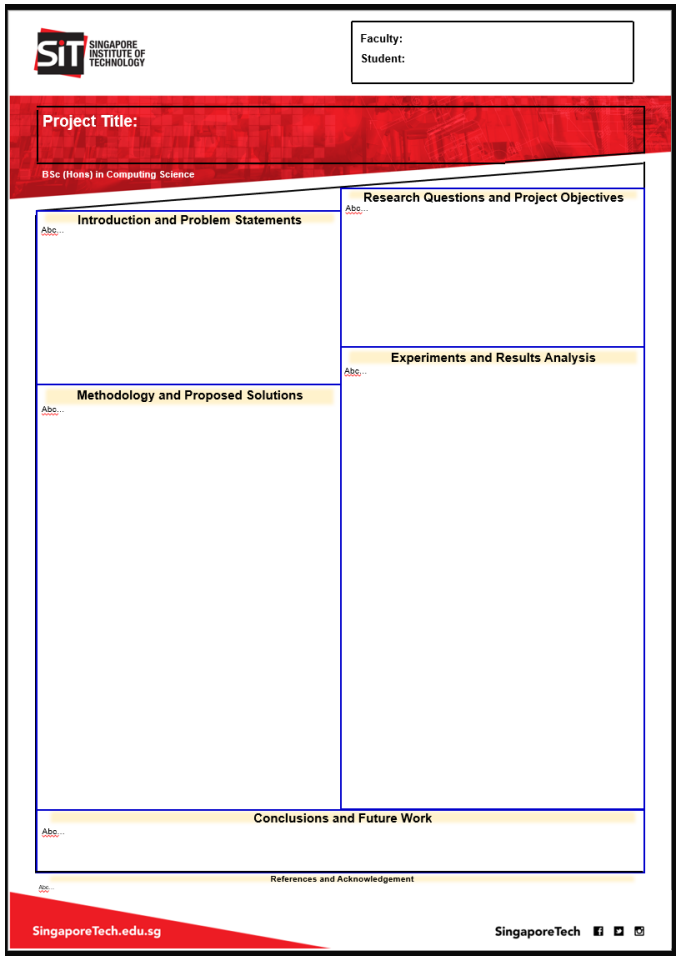
Project Demonstration – Deadline

- Video (10 mins) & Poster **(Week 13: Monday, 30th March @ 0900hrs)**
  - Dropbox at xSITe

- Final Presentation **(Week 13: Thursday, (Scheduled TBD)**
  - 20 mins per Team to Demonstrate project
  - QnA

Project Implementation (30%) - Deliverables

Poster



Video

- **Introduction** (30 seconds)
  Introduce the topic of the projects being showcased. Briefly mention the components and technologies used in the project.
- **Project Overview** (60 seconds)
  Describe the project, including its main features and benefits.
- **Hardware and Software** Components (60 seconds)
  Present a list of the hardware components used in the project, such as microcontrollers, sensors, and actuators. Briefly discuss the software components, including the IoT platform, communication protocols, mobile app, dashboard, etc.
- **Setting up the Edge Devices** (90 seconds)
  Demonstrate the physical setup of the solution, such as connecting the sensors, actuators, and microcontrollers. Briefly explain the configuration of the edge devices, including establishing a connection to the IoT platform.
- **Description of how Edge Devices work** (120 seconds)
  Explain how the edge devices work and interact with various other modules/components. Explain the purpose of the platform and how it enables communication between the devices and the user interface.
- **Edge in Action** (210 seconds)
  Demonstrate how the solution works. Provide examples of relevant scenarios. Show real-time examples of the system in action. Describe the interactions between the devices, edge platform, and mobile app, etc.
- **Conclusion** (30 seconds)
  Summarise the benefits of your projects. Suggest how this project can be used in another context.

Demo

- 20mins
- Demonstrate how it works and its novelty
- QnA

Project Implementation - Rubrics

Rubrics	Description	Excellent	Satisfactory	Bad
Scope Coverage	• How comprehensively does the project cover all specified non-functional requirements? • Are all essential features implemented and functioning as intended?	The project comprehensively addresses all specified non-functional requirements, ensuring all critical features are operational and meet expectations.	While the project covers most of the specified requirements, it may lack some minor functionalities or have incomplete features.	Essential non-functional requirements are not sufficiently addressed, indicating a significant gap in the project's scope coverage.
Integration	• How effectively are the various components/modules integrated with the project? • Is there seamless communication and interaction between different modules?	The project exhibits seamless integration of its components or modules, demonstrating effective communication and synergy between them.	Integration is largely successful, but there are minor issues or limitations in the interaction between different components.	The project shows poor integration, leading to notable communication issues among components.
User Experience	• How intuitive and user-friendly is the user interface? • Does the design prioritise a positive user experience and offer exceptional usability?	The user interface is intuitive and well-designed, ensuring an outstanding user experience and usability.	The user interface is relatively user-friendly, with some aspects needing improvement for better usability and design.	The user interface is not intuitive, leading to significant usability challenges.
Attention to Detail	• How well does the project demonstrate attention to detail, including aspects like responsiveness, consistency, and overall polish?	The project exhibits a meticulous level of detail, including consistency, responsiveness, and overall polish.	Generally, the project is well-executed in terms of detail, but there are minor areas needing refinement.	There is a noticeable lack of detail, resulting in a less polished and refined final product.

Project Implementation – Rubrics

Rubrics	Description	Excellent	Satisfactory	Bad
Scalability & Future-Proofing	• Does the project demonstrate scalability and potential for future enhancements and expansion?	The project demonstrates great scalability, with clear avenues for future enhancements and growth.	The project is somewhat scalable but faces limitations or requires significant redevelopment for expansion.	Scalability is minimal, and major modifications are necessary for any future expansion.
Security and Privacy Measures	• How effectively does the project incorporate security and privacy measures, particularly for IoT devices?	The project incorporates robust security and privacy measures, especially suitable for IoT devices.	Basic security is in place, but there are identifiable vulnerabilities that need addressing.	Security measures are minimal, leaving the project significantly
Testing and Validation	• How comprehensive is the testing and validation process, including considerations for edge cases and stress testing?	Extensive and comprehensive testing, including stress tests and edge case scenarios.	Adequate testing is done but lacks thoroughness or coverage in certain areas.	Testing is minimal, with many areas either untested or only superficially examined.
Interoperability and Standards Compliance	• To what extent is the project interoperable with other systems and compliant with relevant industry standards?	The project is fully interoperable with other systems and complies with all relevant industry standards.	There is general interoperability and compliance but with some exceptions.	The project shows limited interoperability and does not fully comply with industry standards.

Project Guidelines - 1

1. Data Integration from Multiple Sources:
   - Require the design of the project to incorporate data from at least two different sources, such as a camera and a sensor.
   - Emphasize the importance of integrating these diverse data streams effectively, e.g. to reduce the solution complexity.
2. Focus on AI and Machine Learning (ML) Analytics (Emphasis on Edge Analytics):
   - The project should focus on performing analytics at the edge, which involves processing data near its source rather than in a centralised data-processing warehouse.
   - This approach should be practical and applicable in real-world scenarios.
   - Encourage the use of sophisticated ML techniques that are suited for edge devices for data analysis and interpretation.
3. Development of a Dashboard for Data Presentation:
   - Students to create a dashboard that effectively displays raw data and ML analytics results.
   - The dashboard should be user-friendly and capable of illustrating complex data in a comprehensible manner.
   - Emphasis on functionality.
4. Work Package Creation and Individual Responsibilities:
   - Students to divide their project into distinct work packages.
   - Each work package should have a designated individual(s) responsible for its completion.
   - If a work package is handled by two people, it requires a clear division of tasks and responsibilities between them.

Project Guidelines - 2

5. Justification of Hardware and Tools Selection:
   - Students must justify their choice of hardware and tools at each stage of the project. It needs to be captured in documentation in GitHub.
   - Encourage them to consider factors like efficiency, and suitability for the project's objectives.
6. Documentation on GitHub:
   - Require students to maintain thorough documentation of their project on GitHub.
   - This should include detailed explanations of their choices and methodologies, as well as regular updates on project progress.
7. This structured approach will help in ensuring that the students' projects are well-organized, focused on AI and ML, and grounded in practical application, with clear responsibilities and documentation.

Suggested Project Topics

Suggestions

- Smart Health Monitoring Wearable Using Edge IoT and TinyML
- Edge-Based Object Surveillance and Real-Time Tracking System
- Designing a Flexible MCU–FPGA Framework for ML Optimization
- Edge Nodes with Vision, Audio, and Environmental Sensing for Forest Analytics
- HiWonder PuppyPi

Previous Projects

- Voice-Activated Smart Display
- Edge-Powered Avian Deterrence System
- Smart Energy Optimizer
- Enhancing Student learning through Facial Emotional Recognition and/or Voice Tone Analysis in Schools.
- Wearable device to analyze the audience's body language and facial emotions during a presentation *
- Smart Proctoring of Compute Devices
- Canteen Crowd Estimation *

**** You may suggest your own ideas ****

Some Existing Components & Sensors

- ESP32-S3 + 2.4Ghz LoRa
- ESP32 + 923Mhz LoRa
- Pico W *
- M5Stack Plus *
- M5Nano C6
- NRF52840-DONGLE
- RPi 5 *
- NVidia Jetson
  - Nano
  - Xavier
- RPi Zero 2W

- Various Gas Sensors *
- Various Temperature Sensors *
- Ultrasonic Sensors *
- mmWave Sensors
- Webcam *
- **etc**

- LoRaWAN ++ Gateways
- Waveshare Core LR1121 LoRa transceiver *

*** We have many pieces**

Allowable genAI usage, Penalties and Documentation

- You may use genAI tools as an assistant to augment the process of completing your assignment, such as to understand concepts or questions related to assignments; generate ideas for assignments, code and reports; or get suggestions for code improvement, reports, or presentations.
- Remember, genAI is not likely to generate a response that would be seen as quality work and should be modified and improved. It is your responsibility—not the tool's—to ensure the quality, integrity, and accuracy of work you submit.
- You must validate the accuracy of any content, examples and references provided by any genAI tool. For information that are beyond Common Knowledge, you are to find the actual paper(s) or article(s) where the information was likely sourced from, and reference it. This is proper academic practice.

Allowable genAI usage, Penalties and Documentation

- Missing references will be treated as academic dishonesty and penalised as such.
- Should any part of your submission be found to contain any false information, non-existent examples or references, the assignment will be awarded an immediate failure.
- Should a piece of work be suspected to be completed by copying directly from genAI responses, whether in original or paraphrased form, the institution reserves the right to initiate and investigation, as well as call the student for viva voce (oral defense) to ascertain authenticity of the student's work in the submitted assignment.
- Remember to cite the genAI responses using the recommended APA citation format, available at https://libguides.singaporetech.edu.sg/citation/aichatbot